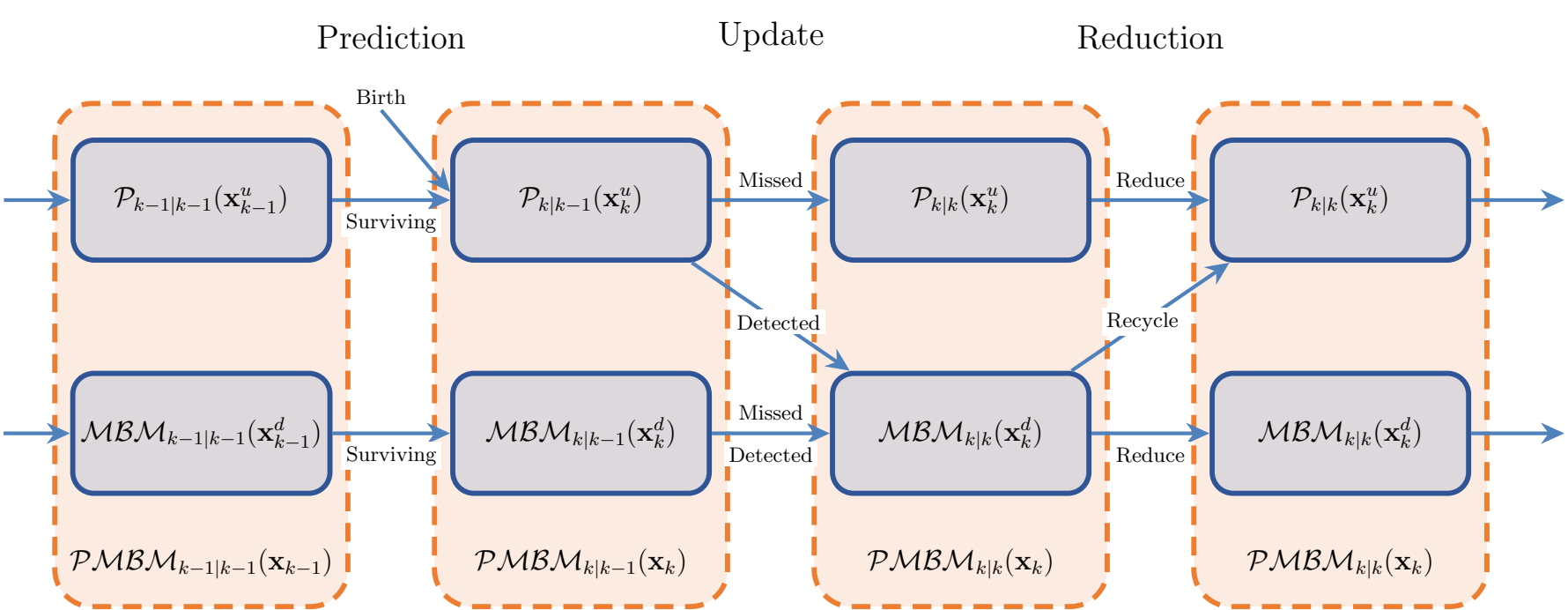

Prediction
Update
Reduction
Birth
$\mathcal{P}_{k-1|k-1}(\mathbf{x}^u_{k-1})$
Surviving
$\mathcal{P}_{k|k-1}(\mathbf{x}^u_k)$
Missed
$\mathcal{P}_{k|k}(\mathbf{x}^u_k)$
Reduce
$\mathcal{P}_{k|k}(\mathbf{x}^u_k)$
Detected
Recycle
$\mathcal{MBM}_{k-1|k-1}(\mathbf{x}^d_{k-1})$
Surviving
$\mathcal{MBM}_{k|k-1}(\mathbf{x}^d_k)$
Missed
Detected
$\mathcal{MBM}_{k|k}(\mathbf{x}^d_k)$
Reduce
$\mathcal{MBM}_{k|k}(\mathbf{x}^d_k)$
$\mathcal{PMBM}_{k-1|k-1}(\mathbf{x}_{k-1})$
$\mathcal{PMBM}_{k|k-1}(\mathbf{x}_k)$
$\mathcal{PMBM}_{k|k}(\mathbf{x}_k)$
$\mathcal{PMBM}_{k|k}(\mathbf{x}_k)$